

Setting	Subset	Stylistic AUROC	SV-Detect AUROC	Gap
<i>MIRAGE-DIG</i>				
DIG	Generate	76.98	99.12	+22.1
DIG	Polish	80.74	95.09	+14.3
DIG	Rewrite	75.94	95.03	+19.1
<i>MIRAGE-SIG</i>				
SIG	Generate	75.59	99.09	+23.5
SIG	Polish	81.71	94.99	+13.3
SIG	Rewrite	76.36	94.21	+17.8
<i>DetectRL (Multi-Attack)</i>				
DetectRL	Direct	90.73	99.99	+9.3
DetectRL	Prompt	90.11	99.96	+9.9
DetectRL	Paraph.	88.69	99.98	+11.3
DetectRL	Perturb.	87.81	99.99	+12.2

Table 3: In-distribution AUROC of the stylistic-feature baseline (Section D.3) compared to the full steering vector pipeline. The **Gap** column shows the additional discriminative signal captured by hidden representations beyond the regex feature bank.

many generators into a single binary detection task and provides an official held-out test set with labels. It therefore offers a complementary large-scale setting for evaluating the same steering vector pipeline and testing its sensitivity to the composition of the generator pool.

E.1 Setup

Data. The COLING-2025 MGT English split contains three partitions: (i) **train**: 610,767 examples (228,922 human, 381,845 machine-generated from 27 generators), (ii) **dev**: 261,758 examples (98,328 human, 163,430 machine-generated), and (iii) **test**: 73,941 examples, released with ground-truth labels for leaderboard evaluation. The task is binary classification (human vs. machine-generated).

Backbones. We evaluate SV-Detect with two reference LMs: the EleutherAI/gpt-neo-2.7B used in the rest of the paper, and the larger meta-llama/Llama-2-7b-hf. For each backbone, we extract per-layer mean-pooled residual activations and use the same downstream logistic-regression detector as in the main paper.

Ablation axes. We vary two components of the pipeline:

1. **Steering vector construction.** We compare the three constructions used elsewhere in the paper: *Mean* (normalized class-mean difference), *PCA* (leading principal component of AI-minus-human activation differences) and

LogReg (ℓ_2 -regularized logistic regression on activations, our default construction).

2. **Generator-pool filter.** The training split contains a long tail of weak open-source generators whose outputs are often much less realistic than those of stronger LLMs. To test whether these generators help or hurt, we compare:

- *all*: all 27 generators
- *trimmed*: remove 20 weak generators (Dolly, BLOOMZ, OPT-*, Flan-T5-*, T0-*), dropping 133,193 samples and retaining 248,652 AI-generated examples. Generators are only removed from the training and dev splits, the whole test split is still used.

We use the same logistic regression detector as described in Sec. 3. The downstream classifier is used elsewhere in the paper. We sweep $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$ and select the best value on dev AUROC. All other hyperparameters match the main experiments as described in Sec. 4.1.

E.2 Results

Llama-based SV-Detect outperforms all baselines by a substantial margin. Removing weak models from the training data helps Llama-based SV-Detect further, gaining about 2 points in Accuracy and 1.7 in F_1 . However, for GPT-Neo-based SV-Detect, removing weak models hurts. We attribute this to the size and capability of the LM backbone: larger and more capable backbones, such as Llama, benefit from a cleaner and more homogeneous generator pool, whereas smaller backbones like GPT-Neo appear to benefit from the additional diversity provided by weaker generators.

E.3 Ablation on steering vector construction

Tab. 5 and Tab. 6 report results for all six combinations of steering vector construction and generator-pool filter on both backbones. As in DetectRL and MIRAGE, the LogReg construction consistently outperforms Mean and PCA. The larger Llama-2-7B backbone further improves substantially over GPT-Neo-2.7B.

Rank	Team / Method	Macro-F1	Acc.
1	SV-Detect (Llama-2-7b, trimmed)	84.8	85.00
2	SV-Detect (Llama-2-7b, all)	83.7	84.00
3	Advacheck	83.07	83.11
4	Unibuc-NLP	83.01	83.33
5	Fraunhofer SIT	82.80	82.89
6	Grape	81.88	82.23
7	TechExperts(IPN)	81.53	81.81
8	TurQUaz	80.68	80.74
9	SzegedAI	79.10	79.29
10	AAIG	78.74	79.34
11	DCBU	77.13	78.01
12	SV-Detect (GPT-Neo, all)	75.6	76.50
13	Alfa	75.37	76.42
14	L3i++	74.63	75.54
15	LuxVeri	74.58	75.68
16	azlearning	74.14	75.17
17	SV-Detect (GPT-Neo, trimmed)	74.0	75.50
18	honghanhh	73.94	75.14
–	Baseline	73.42	74.89
19	VX1291	72.93	74.83
20	cuetttransform	72.32	73.16
21	rockstart	72.24	73.89
22	batirsdu	71.01	71.42
23	IPN-CIC	70.68	72.42
24	Ai-Monitors	70.57	72.65
25	semanticcuet	70.05	71.96
26	hmcgovern	68.48	69.51
27	abhirak0603	68.02	70.50
28	cnlpnitspp	65.02	68.76
29	mail6djj	64.66	68.46
30	bennben	63.32	67.48
31	saehyunma	62.80	67.25
32	yuwert777	62.14	66.69
33	seven	59.09	63.20
34	fangsifan	58.48	62.68
35	yaoxy	57.28	64.20
36	jojoc	54.16	60.37
37	dominikmacko	49.94	50.78
38	tropaleum	49.57	50.60
39	starlight1	47.57	56.65
40	nitstejasrikar	44.89	57.24

Table 4: Adapted English leaderboard from Table 4 of Wang et al. (2025), with our four LogReg-based SV-Detect variants inserted. Ranking is by Macro-F1.

GPT-Neo-2.7B					
SV construction	Pool	Dev AUROC	Test AUROC	Acc.	F_1
Mean	all	0.865	0.860	0.718	0.696
Mean	trimmed	0.855	0.856	0.737	0.721
PCA	all	0.740	0.751	0.559	0.439
PCA	trimmed	0.737	0.753	0.557	0.433
LogReg	all	0.975	0.866	0.765	0.756
LogReg	trimmed	0.952	0.873	0.755	0.740

Table 5: SV-Detect on the COLING-2025 English benchmark with a GPT-Neo-2.7B backbone.

Llama-2-7B					
SV construction	Pool	Dev AUROC	Test AUROC	Acc.	F_1
Mean	all	0.922	0.912	0.716	0.690
Mean	trimmed	0.908	0.887	0.694	0.666
PCA	all	0.864	0.796	0.650	0.615
PCA	trimmed	0.858	0.798	0.647	0.610
LogReg	all	0.984	0.938	0.840	0.837
LogReg	trimmed	0.974	0.937	0.850	0.848

Table 6: SV-Detect on the COLING-2025 English benchmark with a Llama-2-7B backbone.

F Results Tables

In this section, we provide comprehensive tables with results for all experiments in Sec. 4.

F.1 DetectRL

Tables 7 and 8 report the in-domain and cross-source performance of SV-Detect and the baselines, respectively. Fig. 3a, Fig. 3b and Fig. 3c provide a visual summary of these results.

The RepreGuard row of Table 7 is reproduced from the official implementation using our default GPT-Neo-2.7B backbone. For LRR / DetectGPT / Fast-DetectGPT / SV-Detect, we use the training and evaluation protocol described in Sec. 4.1. Results show that RepreGuard with GPT-Neo-2.7B as a backbone collapses to near-chance across almost all cells. Additional comparison of SV-Detect with RepreGuard using LLaMA-3.1-8B as a backbone (default backbone of RepreGuard) is reported in Appendix G.

F.2 MIRAGE

Table 9 reports the performance of SV-Detect and the baselines on the MIRAGE benchmark. Fig. 4 provides a visual summary of these results.

F.3 Ablation

Tables 10 and 11 provide results of ablating the choice of the downstream classifier and steering vector construction method. Sec. 4.3 provides a summary of these results.

F.4 Cross-benchmark transfer: bidirectional MIRAGE ↔ DetectRL

This subsection provides comprehensive results used for Fig. 5 in the cross-benchmark transfer subsection in Sec. 4, broken down by all constituent settings and cells of each target benchmark. We compare SV-Detect against three zero-shot baselines (Fast-DetectGPT, Binoculars, RADAR) that use no training data on either benchmark, and two matched-supervision RoBERTa baselines fine-tuned on the same data as SV-Detect.

On the MIRAGE → DetectRL side (Table 12) we additionally report two multi-direction extensions of SV-Detect. The default variant *polish-500* follows the setup in Sec. 4.1 and is trained on the original 500 human/machine pairs, where machine-generated side is GPT-3.5-Turbo-polished text. The *1-vec (union-800)* extends the training pool to 800 pairs, adding two task-specific subsets

from (Chen et al., 2024): 150 GENERATE pairs, where the machine side is directly generated GPT-3.5-Turbo XSum text, and 150 REWRITE pairs, where the machine side is a GPT-3.5-Turbo rewrite of the corresponding human text. *1-vec (union-800)* keeps the single-direction-per-layer setting, fitting one steering vector per layer on all 800 training pairs. The *3-vec (task-aware)* setting fits a separate steering direction per training subset (POLISH / GENERATE / REWRITE) at each layer and combines them into an orthonormal basis via QR decomposition. The matched-supervision RoBERTa baselines are fine-tuned on the same 500-pair default training set as the default *polish-500* variant. Since Fast-DetectGPT, Binoculars, and RADAR are zero-shot on DetectRL, their transfer numbers coincide with their in-domain scores on the DetectRL test split.

On the DetectRL → MIRAGE side (Table 13) SV-Detect and both RoBERTa baselines are trained on the same joint union of the four DetectRL Multi-Domain training subsets. The task-aware *4-vec* extension learns one steering direction per DetectRL Multi-Domain training subset and, following the extension protocol of MIRAGE, combines them into an orthonormal basis via QR decomposition.

F.5 Cross-benchmark transfer to PADBen and RAID

Table 14 reports per-task AUROC on PADBen’s five sentence-pair regimes alongside the RAID TPR@FPR=5% Chat / Non-Chat / Overall summary. Both benchmarks are used purely as *targets*, and no training data from either is seen by any of the detectors.

The two *default* rows correspond to the default SV-Detect setup described in Sec. 4.1: on MIRAGE, steering vectors are trained on the 500 human/machine pairs, where machine-generated side is GPT-3.5-Turbo-polished text (*MIRAGE polish-500*), and on DetectRL, steering vectors are trained on the joint union of the four Multi-Domain training subsets (*DetectRL 1-vec*). As in Sec. F.4, we additionally report three multi-direction extensions: *MIRAGE 1-vec (union-800)* extends the MIRAGE training pool to 800 pairs as described in Sec. F.4, while keeping a single steering direction per layer; *MIRAGE 3-vec (task-aware)* fits one steering direction per extended MIRAGE training data (Generate / Polish / Rewrite) and combines them via QR decomposition; *DetectRL 4-vec (task-aware)* does the same on the DetectRL side with one direction per Multi-Domain training subset.